Mobile Internet Technology - Formula Sheet

Final revision sheet based on the uploaded MIT lecture notes, assignments, and Q&A; booklet

Topic	Formula	Use it for
Carrier signal	s(t) = A sin(2*pi*f*t + phi)	Identify amplitude A, frequency f, phase phi
Period	T = 1/f	Frequency <-> oscillation duration
Wavelength	lambda = c/f, c ~= 3 x 10^8 m/s	Antenna size / propagation
Bits per symbol	m = log2(M)	PSK/QAM states -> bits per symbol
Nyquist	Rmax = 2 B log2(M)	Maximum rate for ideal/noiseless channel
Nyquist rearranged	B = R / (2 log2(M))	Required bandwidth
Linear SNR	SNR = S/N	Use inside Shannon
SNR in dB	SNRdB = 10 log10(S/N)	Convert linear ratio -> dB
dB to linear	S/N = 10^(SNRdB/10)	Convert dB -> ratio
Shannon	C = B log2(1 + S/N)	Capacity of noisy channel
Required SNR	S/N = 2^(C/B) - 1	Target capacity -> minimum SNR
Free-space received power	Pr = Pt Gt Gr (lambda/(4*pi*d))^2	Propagation / max distance
Path-loss model	Pr proportional to 1/d^alpha	Free space alpha=2; larger in real environments
Propagation delay	t = d/c	One-way signal delay
Round trip time	RTT = 2d/c	Timing advance / RTT
Packet error probability	Ppacket = 1 - (1 - BER)^n	Frame of n bits from BER
Code rate	Rc = k/n	k information bits -> n coded bits
Spreading factor	SF = r_chip / r_data	CDMA / OVSF
CDMA orthogonality	Ci . Cj = 0 (i != j)	Check ideal orthogonal codes
CDMA self-correlation	Ci . Ci = N	Correlation decoding
CDMA superposition	S = sum(Si)	Add chip sequences component-wise
Cell reuse distance	D = R sqrt(3k)	Cell radius R, cluster size k
Transmission time	t = L/R	L bits at rate R
OFDM raw rate	R = Ndata x bits_per_carrier / (Tsymbol + Tguard)	OFDM throughput calculation
WLAN throughput	Throughput = payload bits / total time	MAC efficiency
Backoff duration	Tbackoff = nslots x Tslot	CSMA/CA timeline
ETX	ETX = 1 / (FDR x RDR)	Expected transmissions per link
Path ETX	ETXpath = sum(ETXi)	Compare complete routes
ETT	ETT = ETX x S/B	Expected transmission time
MIMO model	y = sH + n	Transmit vector, channel matrix, noise

Memory Cues - High-Yield Exam Recall

These are the relationships worth recalling before you start any calculation or protocol question.

Modulation & Capacity

- ASK = amplitude; FSK = frequency; PSK = phase; QAM = amplitude + phase.
- Higher-order modulation -> more bits/symbol, but lower robustness and higher SNR requirement.
- Nyquist -> bandwidth + number of signal states.
- Shannon -> bandwidth + noise/SNR.
- Actual achievable rate cannot exceed either the Nyquist or Shannon limit.
- BPSK is more robust than QPSK; QPSK carries more bits per symbol.

Propagation & Multiplexing

- Frequency up -> wavelength down -> physically smaller antenna.
- Received power falls strongly with distance; ideal free space gives 1/d^2.
- Multipath -> delayed copies -> time dispersion -> ISI.
- Protect against ISI with lower symbol rate / guard time / OFDM-style separation.
- Multiplexing dimensions: Time, Frequency, Code, Space.
- FDM guard space = frequency bands; TDM guard space = time; CDMA separation = codes.
- Timing Advance: distant TDMA users transmit earlier to compensate propagation delay.

CDMA / Coding

- CDMA bit mapping used in exercises: 1 -> C, 0 -> -C, idle -> all zeros.
- Orthogonal codes: dot product = 0.
- Spreading factor = chip rate / data rate.
- Near/far problem: strong nearby transmitters overpower weak distant ones; use power control.
- Code rate k/n: fraction of transmitted bits that are information bits.

MAC & WLAN

- Hidden station: transmits when it should not; collision occurs at receiver.
- Exposed station: waits even though it could transmit safely.
- CSMA can still collide because of propagation delay.
- 802.11 uses CSMA/CA, not CSMA/CD.
- SIFS < PIFS < DIFS. Shorter wait = higher priority.
- Backoff freezes when medium becomes busy; remaining backoff is reused.
- RTS -> CTS -> DATA -> ACK; RTS/CTS mitigates hidden stations.
- NAV = virtual carrier sensing / reservation timer.
- DCF = distributed contention; PCF = centralized AP polling.
- Infrastructure: BSS = one AP + stations; ESS = multiple BSSs via DS; IBSS = ad-hoc.

Routing

- DSR + AODV = reactive (on-demand).
- DSDV + OLSR = proactive (routes maintained continuously).
- DSR = source routing; AODV = next-hop table + sequence numbers.
- DSDV = distance vector + sequence numbers; OLSR = link-state optimized with MPRs.
- ETX = expected number of transmissions; lower is better.
- ETT adds transmission time / link rate to ETX.

Mobile Networks & Mobility

- GSM = FDMA + TDMA.
- HSCSD = multiple reserved GSM time slots (circuit-switched).
- GPRS = packet-switched slots on demand.
- EDGE = higher-rate GSM/GPRS via enhanced modulation (notably 8-PSK).
- UMTS = CDMA-based radio access.
- HSPA = HSDPA + HSUPA.
- LTE = OFDM/OFDMA + MIMO + packet core.
- 5G goals: high rate; low latency/high reliability; massive machine communication.
- HLR = home/master subscriber data; VLR = local visitor/location data.
- Mobile IP: MN keeps home address; HA tunnels toward Care-of Address.
- Triangular routing: CN -> HA -> MN, while return may go directly; route optimization can avoid detour.

Final priority: If time is extremely short, memorize Nyquist, Shannon, Friis received power, spreading factor, ETX/ETT, the SIFS-PIFS-DIFS priority order, reactive vs proactive routing, and the GSM to GPRS to UMTS to LTE evolution.